



THIAGO MARZAGAO, Ph.D.

Head of Data Science, Research, and AI at Fitbod

EXPERIENCE

Fitbod

2021 - PRESENT

Head of Data Science, Research, and AI

- **Own the algorithm that generates each workout**, analytics and A/B testing, and applied research. Participate in overall business planning and help build the product roadmap.
- Built Fitbod's experimentation framework, which enabled scaling experimentation from **4 A/B tests in 2021 to 57 in H1 2026**.
- Rewrote the workout algorithm in Python and moved it server-side, enabling **9x more algorithm experiments** and eliminating iOS/Android discrepancies.
- Oversaw algo improvements that reduced exercise replacements/skips by **42%** and increased exact adherence to prescribed sets, reps, and weights from **30% to 44%**.
- Those algo improvements made **strength gains go up by 4x**. Ex. average bench-press strength increase between 1st and 10th sessions went from **6% in 2021 to 26% in 2026**.
- Developed a new One-Rep-Max formula using 303,494 near-failure sets from 14,966 users across 388 exercises; manuscript under review.
- Partnered with leading sports scientists to research the predictors of long-term strength training adherence. Article forthcoming in *Frontiers in Sports and Active Living*.
- Oversaw projects leading to **one granted and one pending U.S. patent**.
- Hired, mentored, and promoted data scientists to senior roles.

Observatory of Public Spending

2016 - 2021

Senior Data Scientist and Deputy Head

- The Observatory of Public Spending pioneered the use of machine learning to find cases of corruption in Brazil. It is part of the Office of the Comptroller-General.
- Trained an ML model that helped the Comptroller-General of Brazil choose which municipalities to audit. **Total auditable budget was US\$ 3.6 billion**.
- Trained an ML model that estimated the price of government-owned real estate properties, to help the government auction them.
- Mined the database of covid economic relief to find cases of fraud.
- Trained and mentored data scientists in other agencies.

Teaching appointments

2016 - 2018

- Taught undergraduate and graduate courses at the University of Brasília (Data Mining), IDP (Introduction to Statistics), and IESB (Data Mining & Machine Learning). Also taught a short course (Machine Learning with Python) at IPEA.

EDUCATION / TRAINING

Ph.D., Quantitative Methods (Department of Political Science) 2010 - 2014

The Ohio State University

- Fulbright scholarship.
- Used machine learning to build the world's first machine-coded democracy index.

Program on Negotiation

2008

Harvard / MIT / Tufts joint program

- Program on Negotiation for Senior Executives
- Dealing with Difficult People and Difficult Situations
- Negotiating Complex Business Deals

CONTACT

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PROFILE

- Data and AI leader with 10+ years across consumer tech, applied research, and fraud detection.
- Ph.D. from a leading U.S. research university.
- Peer-reviewed research in sports science, statistics, economics, and political science.
- Guest on data science podcasts and conferences.

SELECTED CONTENT

- The Brazilian Senate held a hearing on AI & public policy and invited me to talk. [Link](#).
- I was interviewed on the Let's Data podcast. [Link](#).
- I was interviewed on the CTRL+ENTER podcast. [Link](#).
- I gave a talk at the QconSP. [Link](#).
- Paper on predictors of workout adherence, with Federica Conti, Brad Schoenfeld, Andy Galpin. [Link](#).

LANGUAGES

- English (fluent)
- Portuguese (native)
- Spanish (conversational)
- French (basic)